



THE ANATOMY OF REAL NUMBERS



A Masterclass
in Primes,
Proofs, and
Properties

What's inside this chapter?

Part 1: The DNA of Numbers.

(Fundamental Theorem
of Arithmetic)



Part 2: Family Ties.

(HCF & LCM)



Part 3: The Rule Breakers.

(Proving Irrationality)



THE FUNDAMENTAL THEOREM OF ARITHMETIC

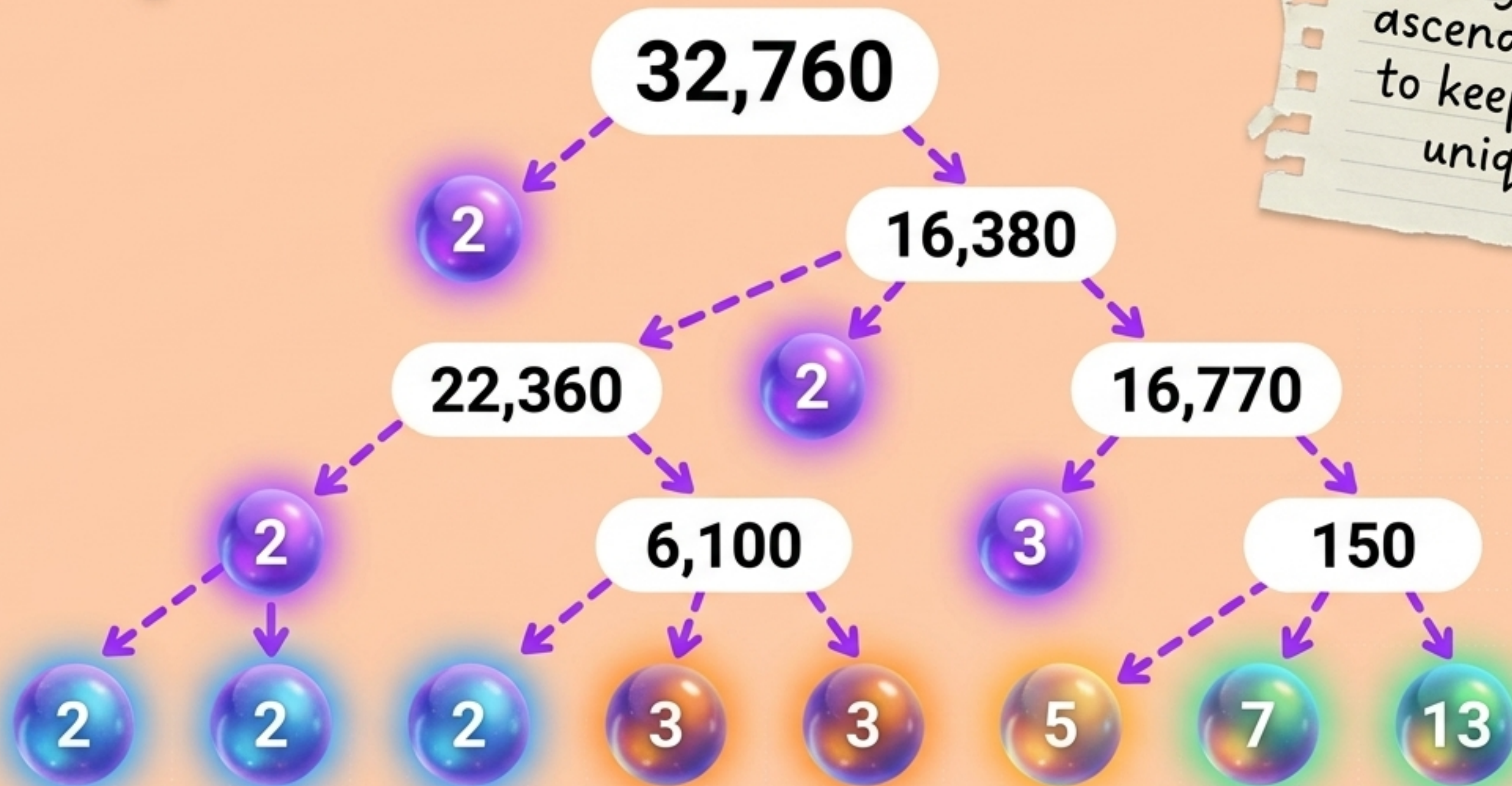
Every **composite number** can be expressed as a product of primes.
This **factorisation** is **UNIQUE** (apart from the order of factors).



Historical Tea ☕:
Carl Friedrich Gauss
(The 'Prince of
Mathematicians') gave
the first correct proof
of this in his book
Disquisitiones
Arithmeticae!

Extracting the DNA: The Factor Tree

Always write in ascending order to keep it 100% unique! 🚀



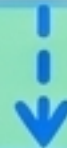
➡ $32760 = 2^3 \times 3^2 \times 5^1 \times 7^1 \times 13^1$

Case Study: The Mystery of the Zero

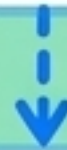
Question: Can the number 4^n ever end with the digit 0?



To end in zero, a number **MUST** have the prime factors **2 AND 5**.



$$4^n = (2^2)^n = 2^{(2n)}$$



Because of the Fundamental Theorem, the **ONLY** prime factor of 4^n is **2**. The prime **5** is missing!



~~5~~ Verdict: **IMPOSSIBLE.**

HCF vs. LCM: The Ultimate Cheat Sheet

Highest Common Factor

Product of the **SMALLEST**
power of each **COMMON** prime.



Lowest Common Multiple

Product of the **GREATEST**
power of **EVERY** prime involved.



Let's see them
in action... →

Family Ties: 6 & 20

The DNA Breakdown

$$6 = 2^1 \times 3^1$$

$$20 = 2^2 \times 5^1$$

The Selection Process - Side-by-Side

HCF Box

Common prime = **2**.
Smallest power = **2^1** .

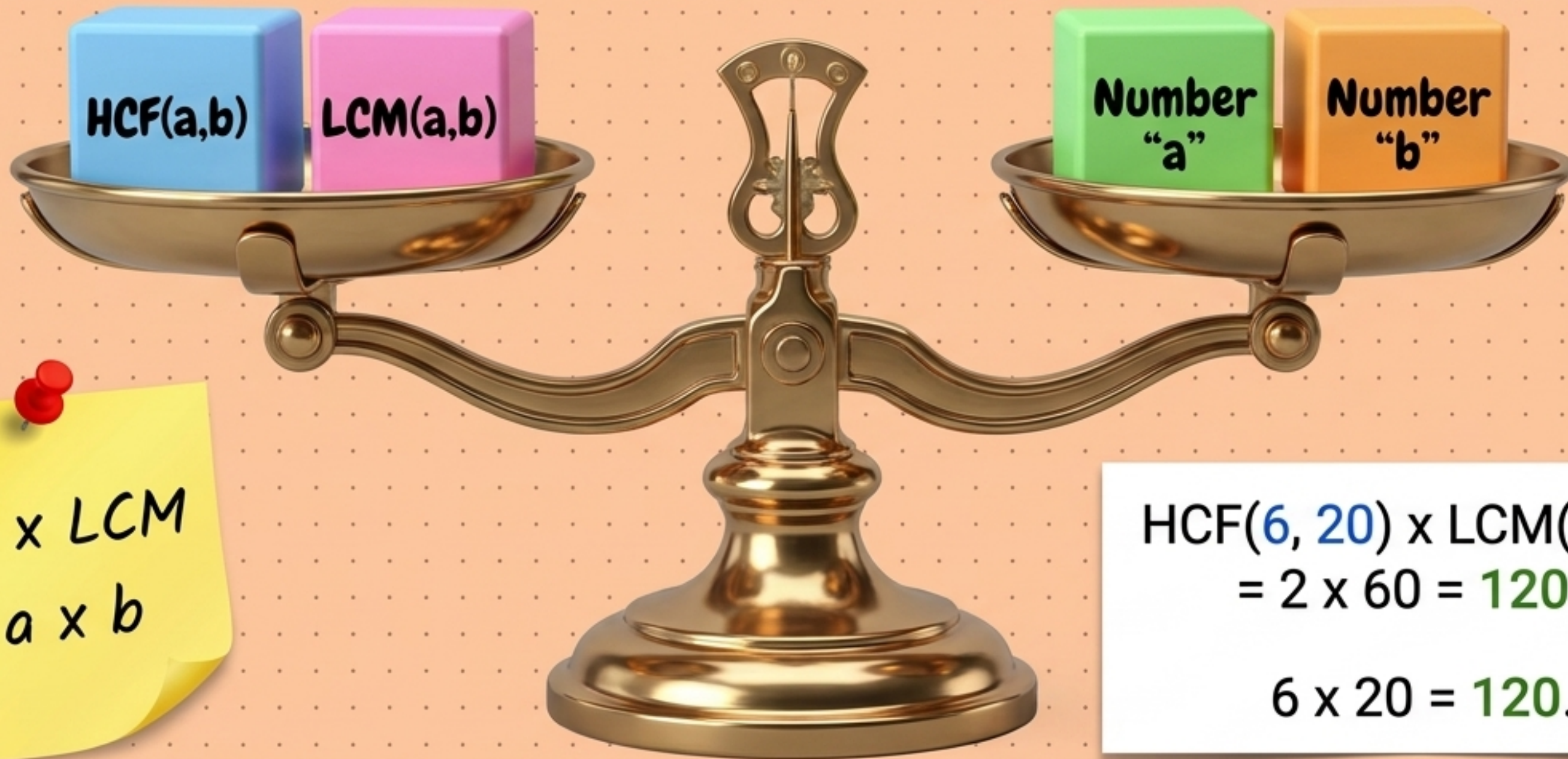
$$\text{HCF} = 2$$

LCM Box

All primes = **2, 3, 5**.
Greatest powers = **$2^2, 3^1, 5^1$** .

$$\text{LCM} = 4 \times 3 \times 5 = 60$$

The Golden Balance Rule



$$\text{HCF} \times \text{LCM} = a \times b$$

$$\begin{aligned} \text{HCF}(6, 20) \times \text{LCM}(6, 20) &= 2 \times 60 = 120. \\ 6 \times 20 &= 120. \end{aligned}$$



Pro-tip: If you know 3 of these variables, you can instantly find the 4th!

THE TRAP: Three's a Crowd!

$$\text{HCF}(p, q, r) \times \text{LCM}(p, q, r) = p \times q \times r$$

Example: 6, 72, 120

HCF = 6, LCM = 360

$6 \times 360 = 2160$

$6 \times 72 \times 120 = 51840$

2160 is NOT EQUAL to 51840! 🙄

Save this for exams! 📌


$$\text{LCM}(p, q, r) = \frac{(p \times q \times r \times \text{HCF}(p, q, r))}{\text{HCF}(p, q) \times \text{HCF}(q, r) \times \text{HCF}(p, r)}$$

Part 3: Catching the Rule Breakers



Theorem 1.2: Let p be a prime number. If p divides a^2 , then p **MUST** also divide a .



Quick Check: If 3 divides 81 (which is 9^2), then 3 definitely divides 9. Make sense? 🧠✨

We need this key for the ultimate math boss fight...

The Detective Method: Proof by Contradiction

Step 1: The Setup

Assume the suspect is INNOCENT (Assume the number is RATIONAL).

Step 2: The Interrogation

Follow the strict rules of algebra and math.

Step 4: The Verdict

Since they lied, our assumption was wrong. They are GUILTY (IRRATIONAL!).

Step 3: The Trap

Watch them break a fundamental rule! (The Contradiction 🚨).



CASE FILE: PROVE $\sqrt{2}$ IS IRRATIONAL

1. The False Assumption

Assume $\sqrt{2}$ is rational. So, $\sqrt{2} = a / b$.

CRUCIAL DETAIL: Assume a and b are **COPRIME** (they share **NO** common factors other than 1).

2. The Algebra

Rearrange: $b\sqrt{2} = a$

Square both sides: $2b^2 = a^2$



This means a^2 is divisible by 2. By our Golden Key (Thm 1.2), a is divisible by 2!

The Trap Springs!

Step 3: Substitute

Since a is divisible by 2, we can write: $a = 2c$

Step 4: The Switch

Substitute ' $2c$ ' back into the equation $2b^2 = a^2$:

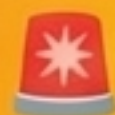
$$2b^2 = (2c)^2$$

$$2b^2 = 4c^2$$

$$b^2 = 2c^2$$

Step 5: The Realization

Wait! This means b^2 is divisible by 2. So b is divisible by 2!



OBJECTION! We said ' a ' and ' b ' had NO common factors. But they are BOTH divisible by 2!
The assumption is destroyed. $\sqrt{2}$ is IRRATIONAL.

The Blueprint Works for $\sqrt{3}$ Too!

1. The False Assumption

Assume $\sqrt{2}$ is rational. So, $\sqrt{2} = a/b$.

CRUCIAL DETAIL: Assume a and b are **COPRIME** (they share **NO** common factors other than 1).

2. The Algebra

Rearrange: $b\sqrt{2} = a$

Step 4: The Switch

Substitute ' $2c$ ' back into the equation $2b^2 = a^2$:

$$2b^2 = (2c)^2$$

$$2b^2 = 4c^2$$

$$b^2 = 2c^2$$

Step 5: The Realization

Wait! This means b^2 is divisible by 2. So b is divisible by 2!

OBJECTION! We said ' a ' and ' b ' had **NO** common factors. But they are **BOTH** divisible by 2! The assumption is destroyed. $\sqrt{2}$ is **IRRATIONAL**.

Assume $\sqrt{3} = a/b$ (coprime).

$3b^2 = a^2 \rightarrow a$ is divisible by 3.

Substitute $a = 3c \rightarrow 3b^2 = 9c^2$
 $\rightarrow b^2 = 3c^2$.

b is divisible by 3.

Notice the pattern?
You just swap the 2s
for 3s. The logic
trap is exactly the
same! 🧩

The Copycats: Proving Combinations

When you mix a rational and an irrational, the result is irrational. The proof is much shorter!

Prove $5 - \sqrt{3}$ is irrational.

Assume rational: $5 - \sqrt{3} = a/b$

Rearrange to isolate $\sqrt{3}$:
 $\sqrt{3} = 5 - a/b$

★ **TRAP:** The right side is Rational, but we KNOW $\sqrt{3}$ is Irrational. Contradiction!

Sum / Difference

Prove $3\sqrt{2}$ is irrational.

Assume rational: $3\sqrt{2} = a/b$

Isolate $\sqrt{2}$: $\sqrt{2} = a/(3b)$

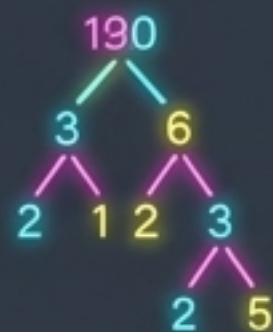
★ **TRAP:** Right side is Rational, left is Irrational. Boom.

Product / Quotient

The Ultimate Real Numbers Cheat Sheet

The DNA

Fundamental Theorem:
Every number has unique
prime DNA.



The Family

HCF = Smallest powers of common primes
LCM = Greatest powers of all primes
 $\text{HCF} \times \text{LCM} = a \times b$



The Rule Breakers

Proof by Contradiction Blueprint:
Assume Rational (a/b coprime) $\rightarrow$ Do Algebra $\rightarrow$ Prove they share a factor $\rightarrow$ Trap! It's Irrational.

Save this to your study board!  